

Exercises, sheet 4. In the following exercises sometimes a theory is mentioned but not precisely defined (e.g. the theory of algebraically closed fields). It is then the task of the reader to axiomatize it in some language in the first-order way.

1. Let M be an infinite countable L -structure, where L is countable. Prove that $Th(M)$ is ω -categorical if and only if $\text{Aut}(M)$ has only finitely many orbits when acting on M^n for any natural n .
2. Show that a theory T eliminates quantifiers if and only if every formula of the form $\exists x \psi(x, \bar{y})$ is T -equivalent to a quantifier-free formula. Deduce the following criterion for quantifier elimination:
 (★) For every L -structure A and embeddings $A \subseteq M$, $A \subseteq N$ into κ -saturated models $M, N \models T$, for every $a \in M$, there is $b \in N$ such that $\text{qftp}(a/A) = \text{qftp}(b/A)$.
 By $\text{qftp}(a/A)$ we mean the collection of quantifier-free formulas in $L(A)$ satisfied by a in M (similarly with $\text{qftp}(b/A)$).
3. Prove that the theory of vector spaces over a fixed field K eliminates quantifiers. Characterise its completions depending on K .
4. Prove that the theory ACF (algebraically closed fields) eliminates quantifiers. Prove that the characteristic p or 0 characterises its completions.
5. Show that if F is the algebraic closure of a finite field and $f : F^n \rightarrow F^n$ is a function that whose each coordinate is a polynomial, then injectivity of f implies surjectivity of f . Hint: look at intermediate finite fields.
6. Fix a degree d and a number of variables n . Note that the statement “every map $f : F^n \rightarrow F^n$ whose entries are polynomials of degree less or equal d that is injective is also surjective” can be encoded by a first-order sentence σ in the ring language, about the field F . Show that $\text{ACF}_0 \vdash \sigma$ if and only if $\text{ACF}_p \vdash \sigma$ is valid for big enough p . Conclude that $\mathbb{C} \models \sigma$, i.e., with the Ax-Grothendieck theorem.
7. An element $b \in B$ in a Boolean algebra B is called an *atom* if for every $b' \subset b$ (i.e., $\neg b' \vee b = 1$) we have $b' = 0$ or $b' = b$. Show that the theory of atomless Boolean algebras eliminates quantifiers, is complete, and ω -categorical.
8. Show that theories considered in the previous exercises are model companions of their universal theories (identify those).